Experiment No. 3

```
# Import the required libraries
import pandas as pd
import numpy as np
import sklearn
from sklearn import datasets
```

```
digit=datasets.load_digits()
digit
```

```
{'data': array([[ 0.,  0.,  5., ...,  0.,  0.,  0.],
        [ 0.,  0.,  0., ..., 10.,  0.,  0.],
        [ 0.,  0.,  0., ..., 16.,  9.,  0.],
        ...,
        [ 0.,  0.,  1., ...,  6.,  0.,  0.],
        [ 0.,  0.,  2., ..., 12.,  0.,  0.],
        [ 0.,  0., 10., ..., 12.,  1.,  0.]]),
 'target': array([0, 1, 2, ..., 8, 9, 8]),
 'frame': None,
 'feature_names': ['pixel_0_0',
  'pixel_0_1',
  'pixel_0_2',
  'pixel_0_3',
  'pixel_0_4',
  'pixel_0_5',
  'pixel_0_6',
  'pixel_0_7',
  'pixel_1_0',
  'pixel_1_1',
  'pixel_1_2',
  'pixel_1_3',
  'pixel_1_4',
  'pixel_1_5',
  'pixel_1_6',
  'pixel_1_7',
  'pixel_2_0',
  'pixel_2_1',
  'pixel_2_2',
  'pixel_2_3',
  'pixel_2_4',
  'pixel_2_5',
  'pixel_2_6',
  'pixel_2_7',
  'pixel_3_0',
  'pixel_3_1',
  'pixel_3_2',
  'pixel_3_3',
  'pixel_3_4',
  'pixel_3_5',
  'pixel_3_6',
  'pixel_3_7',
  'pixel_4_0',
  'pixel_4_1',
  'pixel_4_2',
  'pixel_4_3',
  'pixel_4_4',
  'pixel_4_5',
  'pixel_4_6',
  'pixel_4_7',
  'pixel_5_0',
  'pixel_5_1',
  'pixel_5_2',
  'pixel_5_3',
  'pixel_5_4',
  'pixel_5_5',
  'pixel_5_6',
  'pixel_5_7',
```

```
df = pd.DataFrame(digit['data'])
df.head(5)
```

	0	1	2	3	4	5	6	7	8	9	...	54	55	56	57	58	5
0	0.0	0.0	5.0	13.0	9.0	1.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	6.0	13.(
1	0.0	0.0	0.0	12.0	13.0	5.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	11.(
2	0.0	0.0	0.0	4.0	15.0	12.0	0.0	0.0	0.0	0.0	...	5.0	0.0	0.0	0.0	0.0	3.(
3	0.0	0.0	7.0	15.0	13.0	1.0	0.0	0.0	0.0	8.0	...	9.0	0.0	0.0	0.0	7.0	13.(
4	0.0	0.0	0.0	1.0	11.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	2.(

5 rows × 64 columns

```
df[4]=digit['target']
df.head(4)
```

	0	1	2	3	4	5	6	7	8	9	...	54	55	56	57	58	59
0	0.0	0.0	5.0	13.0	0	1.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	6.0	13.0
1	0.0	0.0	0.0	12.0	1	5.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	11.0
2	0.0	0.0	0.0	4.0	2	12.0	0.0	0.0	0.0	0.0	...	5.0	0.0	0.0	0.0	0.0	3.0
3	0.0	0.0	7.0	15.0	3	1.0	0.0	0.0	0.0	8.0	...	9.0	0.0	0.0	0.0	7.0	13.0

4 rows × 64 columns

```
# Keep only first 4 columns dynamically
df = df[df.columns[:4]]

df.head()
```

	0	1	2	3
0	0.0	0.0	5.0	13.0
1	0.0	0.0	0.0	12.0
2	0.0	0.0	0.0	4.0
3	0.0	0.0	7.0	15.0
4	0.0	0.0	0.0	1.0

Next steps: Generate code with df | New interactive sheet

```
#Adding cloumns names
df.rename(columns={0:'pixel 1' , 1:'pixel 2' , 2:'pixel 3' , 3:'pixel 4' }  , i
df.head(4)
```

	pixel 1	pixel 2	pixel 3	pixel 4
0	0.0	0.0	5.0	13.0
1	0.0	0.0	0.0	12.0
2	0.0	0.0	0.0	4.0
3	0.0	0.0	7.0	15.0

Next steps: Generate code with df | New interactive sheet

```
df.describe()
```

	pixel 1	pixel 2	pixel 3	pixel 4
count	1797.0	1797.000000	1797.000000	1797.000000
mean	0.0	0.303840	5.204786	11.835838
std	0.0	0.907192	4.754826	4.248842
min	0.0	0.000000	0.000000	0.000000
25%	0.0	0.000000	1.000000	10.000000
50%	0.0	0.000000	4.000000	13.000000
75%	0.0	0.000000	9.000000	15.000000
max	0.0	8.000000	16.000000	16.000000

```
df.shape
```

```
(1797, 4)
```

MEAN

```
df.mean()
```

	0
pixel 1	0.000000
pixel 2	0.303840
pixel 3	5.204786
pixel 4	11.835838

dtype: float64

MEDIAN

```
df.median()
```

	0
pixel 1	0.0
pixel 2	0.0
pixel 3	4.0
pixel 4	13.0

dtype: float64

MODE

```
# Calculated only for categorical data
df['pixel 4'].mode()
```

	pixel 4
0	16.0

dtype: float64

```
df.groupby(['pixel 2']).count()
```

pixel 2	pixel 1	pixel 3	pixel 4
0.0	1531	1531	1531
1.0	128	128	128
2.0	67	67	67
3.0	32	32	32
4.0	22	22	22
5.0	9	9	9
6.0	3	3	3
7.0	3	3	3
8.0	2	2	2

STANDARD DEVIATION

```
df['pixel 1'].std()
```

```
0.0
```

```
df['pixel 2'].std()
```

```
0.907192095250743
```

```
df['pixel 3'].std()
```

```
4.754826339660716
```

```
df['pixel 4'].std()
```

```
4.248841848260788
```